

Medical DB Query Generation for SR

Currently: Seed papers $\Rightarrow$ Key terms $\longrightarrow$ come up with query
use strategies, combination of keywords & their variations
 $\longleftarrow$ trial & error
w/ multiple queries (combination of a bunch)

Goal: $\longrightarrow$ get concise studies, yet don't overfetch 

Initial Approach:

Step 1) Get keywords $\longrightarrow$ SR into LLM to get PICO
 $\hookrightarrow$ Summarize $\longrightarrow$ TFID or some kind of similarity word search

```
1 "Myocardial Ischemia"[Mesh] OR "Coronary Artery Disease"[Mesh] OR
  "Acute Coronary Syndrome"[Mesh] OR "Myocardial Infarction"[Mesh] OR
  "Angina Pectoris"[Mesh] OR "Percutaneous Coronary Intervention"[Mesh]
  OR "Coronary Artery Bypass"[Mesh]

2 Myocardial Ischemia[Title/Abstract] OR Ischemic Heart
  Disease[Title/Abstract] OR Heart Disease*[Title/Abstract] OR Heart
  attack[Title/Abstract] OR Coronary Heart Disease[Title/Abstract] OR
  Coronary Artery Disease[Title/Abstract] OR Coronary Disease*[Title/Ab-
  stract] OR Acute Coronary Syndrome[Title/Abstract] OR Acute Coronary
  Syndrom*[Title/Abstract] OR Myocardial Infarction[Title/Abstract] OR
  Angina*[Title/Abstract] OR Percutaneous Coronary
  Intervention[Title/Abstract] OR Percutaneous Coronary Revascularization*
  [Title/Abstract] OR Percutaneous Transluminal Coronary Angioplast*
  [Title/Abstract] OR Coronary Balloon Angioplast*[Title/Abstract] OR
  Coronary Artery Bypass*[Title/Abstract]

3 #1 OR #2

4 ("Rehabilitation"[Mesh]) OR (Rehabilitation[Title/Abstract] OR
  rehab[Title/Abstract])

5 "Telemedicine"[Mesh] OR mhealth[Title/Abstract] OR m-health[Title/Ab-
  stract] OR mobile health[Title/Abstract] OR mobile application[Title/Ab-
  stract] OR mobile device[Title/Abstract] OR mobile communication*
  [Title/Abstract] OR mobile phone[Title/Abstract] OR smartphone[Title/Ab-
  stract] OR smart phone[Title/Abstract] OR smartphone application[Title/Ab-
  stract] OR cellphone[Title/Abstract] OR cellular phone[Title/Abstract] OR
  telemedicine[Title/Abstract] OR telerehabilitation[Title/Abstract] OR tele-re-
  habilitation[Title/Abstract] OR Virtual Rehabilitation[Title/Abstract] OR
  Remote Rehabilitation[Title/Abstract] OR telehealth[Title/Abstract] OR tele-
  health[Title/Abstract] OR telemonitor[Title/Abstract] OR ehealth[Title/Ab-
  stract] OR e-health[Title/Abstract] OR digital health[Title/Abstract] OR mo-
  bile technolog*[Title/Abstract] OR website[Title/Abstract] OR web-
  based[Title/Abstract]
```

6 3 and 4 and 5

Step 2) Make queries.
 $\Rightarrow$ Common method:

bunch of **OR** w/ initial searches
 $\rightarrow$ then aggregate everything with
AND from previous results.

Step 3)

Approach 2) Include studies in bibliography. w/ seed papers.

Step 1) Fetch all papers referenced, even some not in "include.xlsx" & parse them

Step 2) Build PICO for SR itself & compare relevance against all references using a similarity model. Filter all studies below certain threshold.

Step 3) W/ Filtered studies parse their abstracts if available & get relevant terms.

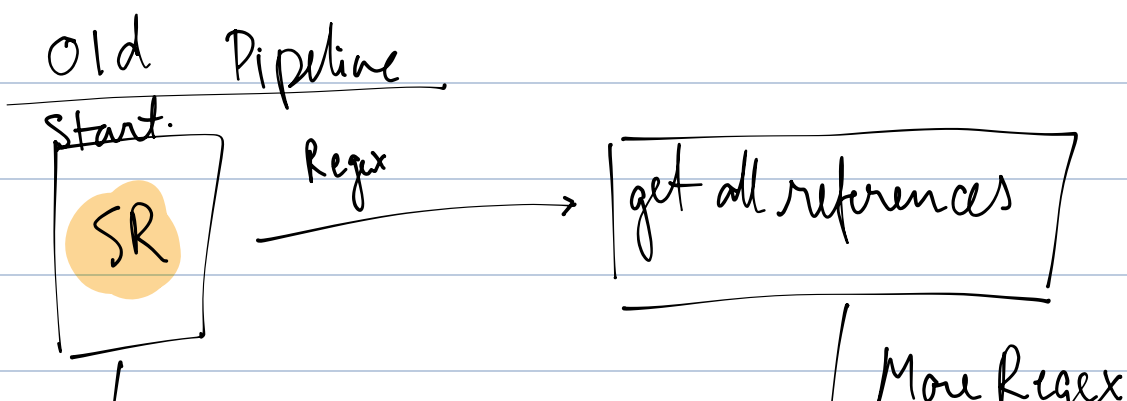
from pico → look for terms

⇒ Insight: 1) To get sensitive list: use precise terms
e.g. renal failure vs kidney

2) Get good accuracy / good recall. → include all term synonyms
Overall → include all synonym terms within PICO and be precise. no vague term (they give the shit results)

Article I found on LLMs making queries:

<https://academic.oup.com/jamiaopen/article/7/3/ooae098/7775509>



LLM

Get title or DOI if available

Get PICO Facets
→ Mainly population & Intervention

Run queries on PubMed AP

Reference Metadata
(Not so good: has a lot of missing return values)

TF-IDF

Gemini

→ cached into JSON file

Cosine similarity search to root out useless references:
Final relevant text

Gemini

Extract Keywords either MESH or free text.

From PICO, Gem. extracts n key words from population & interventions (e.g. 3)

Gemini

try to filter

Extracted terms get refined to make more relevant to "Core idea" with keywords

with final filtered words

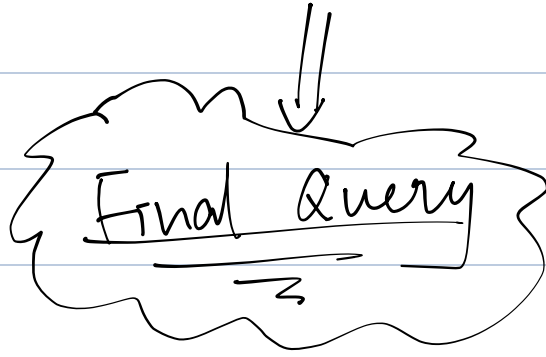
Gemini

→ make query: mix of Mesh and free text

ex: (pop_1 or pop_2 or) and (inter_1 or inter_2 or ...) and RCT

cuts results by a lot

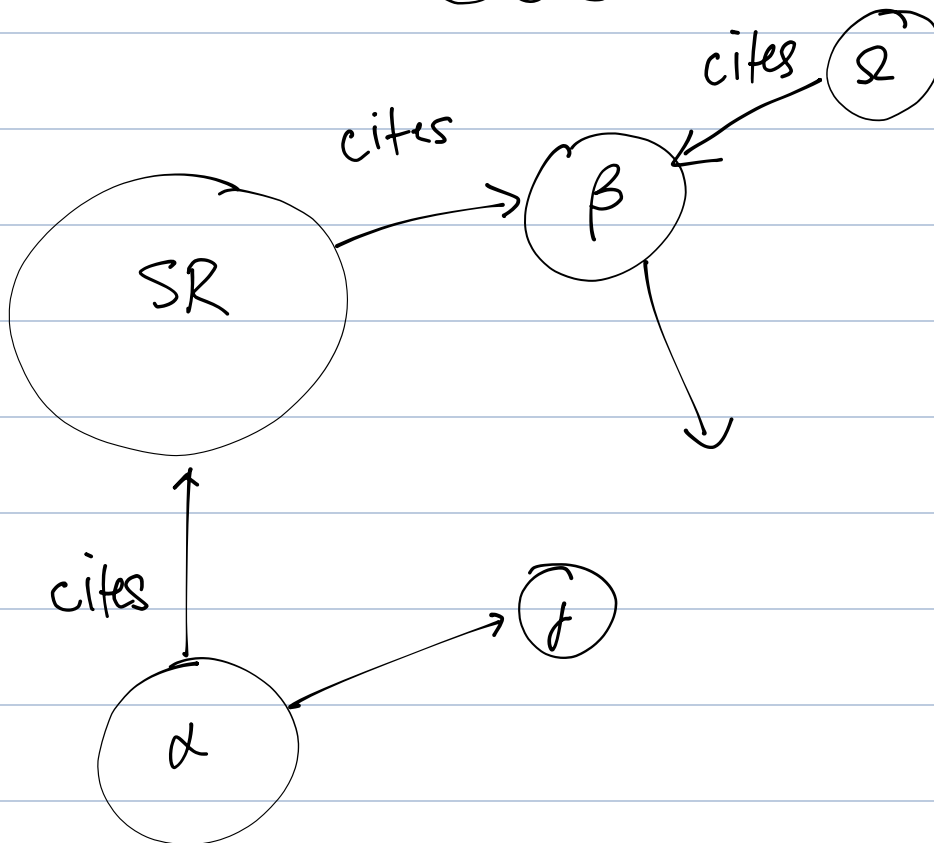
} randomized control trial
(if SR - deems it so)



Observations: used gemini because of free credits.

⇒ results vary, good ones include ~ 900 results for 88% recall

New Approach (Graph Method)



Final result is big as adjacency list to better

represent connections between article = better gauge for relevancy + covers more literature

↳ Issues w/ previous method:

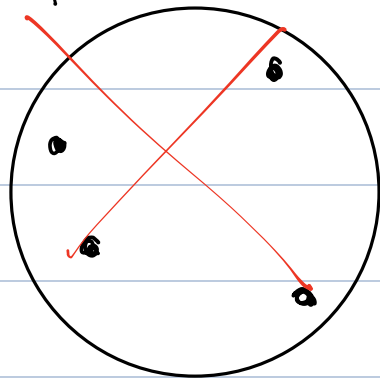
1) - Too precise or too vague. → filtering wrong words

2) → some articles don't have MESH

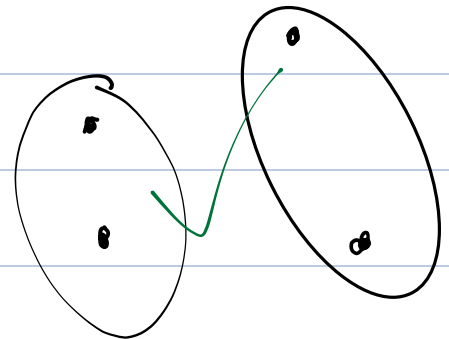
→ Exceptions for lack of recall, some articles not in PubMed, some DOI don't work, etc...

⇒ **Solution:** More context: BFS to make adjacency list w/ citations + adding more variant to core words.

Result: Get more specific terms & their variant so



instead



Relying on Mesh words

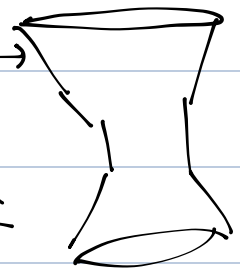
less noise with free words.

Insights: → Discovered that MeSH terms are very unpredictable, slight change can lead to amount of results to explode. ⇒ Conclusion: Make MESH terms quite specific in query and add more variants and more synonyms in free terms.

⇒ Basic Idea: Get as much context as possible, filter ideas, then broaden terms

Final Pipeline

like "Double funnel" →



Start.
SR seed
papers

Build citation Graph

LLM

Prospero

✓ hop-0 N-random seed articles from rbc

X Discarded

hop-1: articles that cite the N seeds

✓ top30: hop-2: All references of hop-1, aka.
all cocitations of initial seeds

✓ top10: hop-3: Articles cited/reference hop-2

Get PICO
Facets
→ Mainly population
& Intervention

Mesh extraction

Free-text attraction

Hop 0

Gemini splits into pop/intervention

Hop 2/3

Gemini takes top 10 most
relevant for Pop/intervention

} Pooled into set
then split together

Prospero

Gem splits. independently → same.

In the end:

Pop Mesh: [...].

Intervention Mesh [...].

Pop Free-text: [...].

Intervention Free-text [...].

Wild Card expansion: add synonyms + * wildcards

Final Cleaning for too broad or
irrelevant terms

Query Builder: (Pop terms or) AND (Intervention Terms)

Trial & Error Observations (No particular order)

- 1) Prospero content is REALLY Relevant, so skips cleaning
- 2) LLMs are unpredictable, for every API call we send identical one the combine results e.g. call 1: A B, call 2: AC
→ final answer: ABC
- 3) Demographic info always bloats results, so HARD BAN
- 4) For more info, we analyse the seed titles as baseline.
- 5) Population Mesh terms are pretty concise usually so we skip augmentation w/ hop 2-3, we stop at hop 0 + prospero.
- 6) Broad terms are usually 1 word, so all one-word terms are removed. → not always but in general, for precision

Challenges: LLMs in general

- Often too harsh or too loose on term extraction or cleaning so a lot of prompt tweaks.
- doesn't follow instructions (typical)
- Case per case testing: some SR
- Graph building takes a lot of time from fetching/counting references, but results are cached in an adjacency list.
- Long list terms take a long time to filter and cause

mistakes.

- Average time 30h min + graph building → 6-7min
- On occasion, returns very big n° of results >100k
 - ↳ when topics are pretty vague, and LLMs can't differentiate between relevant & irrelevant (going back to case per case testing)